

AMERICAN WELDING SOCIETY

Basic Welding Symbols and Their Location Significance

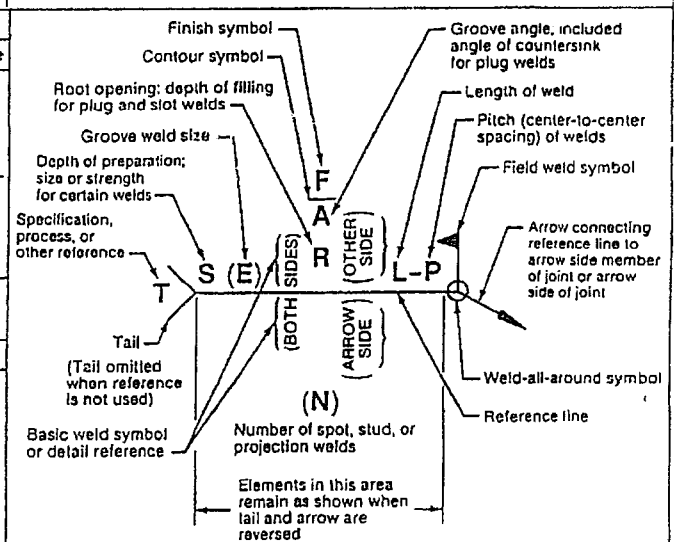
Location Significance	Fillet	Plug or Slot	Spot or Projection	Stud	Seam	Back or Backing	Surfacing	Flange Corner	Flange Edge
Arrow Side									
Other Side				Not used			Not used		
Both Sides		Not used	Not used	Not used	Not used	Not used	Not used	Not used	Not used
No Arrow Side or Other Side Significance	Not used	Not used		Not used		Not used	Not used	Not used	Not used

Location Significance	Groove							Scarf for Brazed Joint
	Square	V	Bevel	U	J	Flare-V	Flare-Bevel	
Arrow Side								
Other Side								
Both Sides								
No Arrow Side or Other Side Significance		Not used	Not used	Not used	Not used	Not used	Not used	Not used

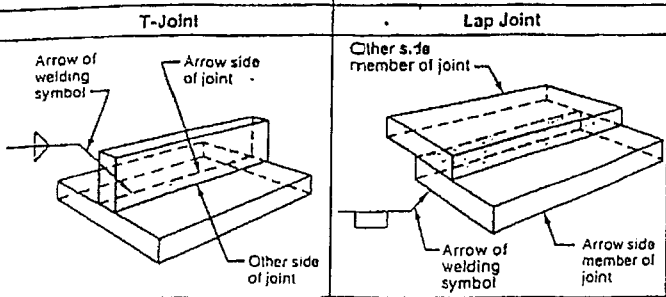
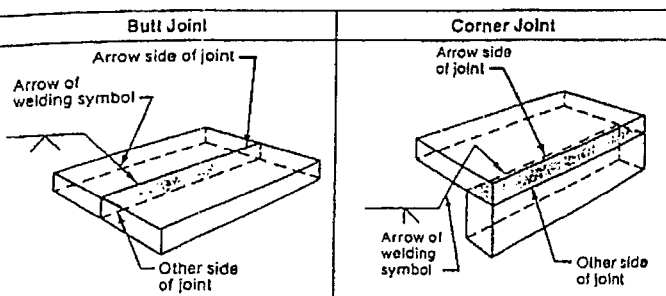
Supplementary Symbols

Weld-All Around	Field Weld	Melt-Thru	Consumable Insert	Backing Spacer	Contour		
					Flush	Convex	Concave

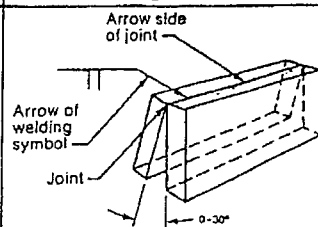
Location of Elements of a Welding Symbol



Basic Joints
Identification of Arrow Side and Other Side of Joint



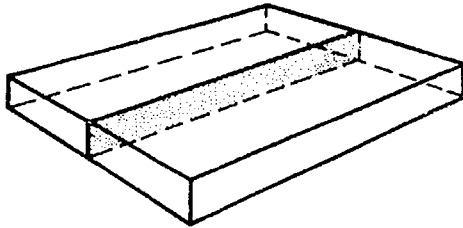
Edge Joint



Process Abbreviations

Where process abbreviations are to be included in the tail of the welding symbol, reference is made to Table 1, Designation of Welding and Allied Processes by Letters, of AWS A2.4-86

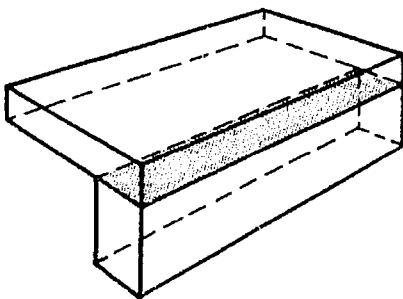
AMERICAN WELDING SOCIETY, INC.
550 N.W. LeJeune Rd., Miami, Florida 33126



(A) Butt joint

Applicable welds

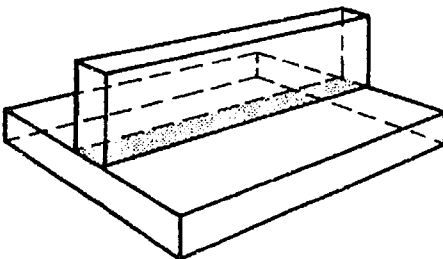
Bevel-groove	U-groove
Flare-bevel-groove	V-groove
Flare-V-groove	Edge-flange
J-groove	Braze
Square-groove	



(B) Corner joint

Applicable welds

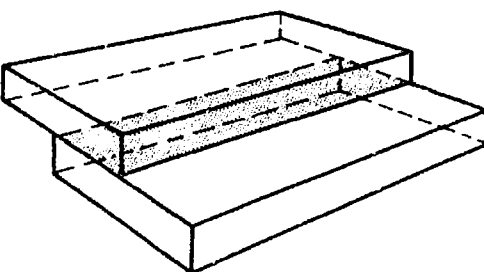
Fillet	Corner-flange
Bevel-groove	Edge-flange
Flare-bevel-groove	Plug
Flare-V-groove	Slot
J-groove	Spot
Square-groove	Seam
U-groove	Projection
V-groove	Braze



(C) T-joint

Applicable welds

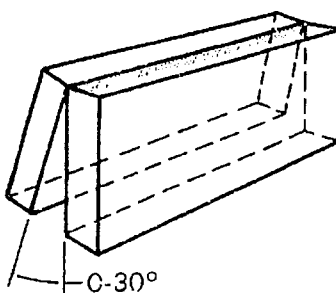
Fillet	Slot
Bevel-groove	Spot
Flare-bevel-groove	Seam
J-groove	Projection
Square-groove	Braze
Plug	



(D) Lap joint

Applicable welds

Fillet	Slot
Bevel-groove	Spot
Flare-bevel-groove	Seam
J-groove	Projection
Plug	Braze



(E) Edge joint

Applicable welds

Bevel-groove	V-groove
Flare-bevel-groove	Edge
Flare-V-groove	Corner-flange
J-groove	Edge-flange
Square-groove	Seam
U-groove	

Fig. 1—Joint types

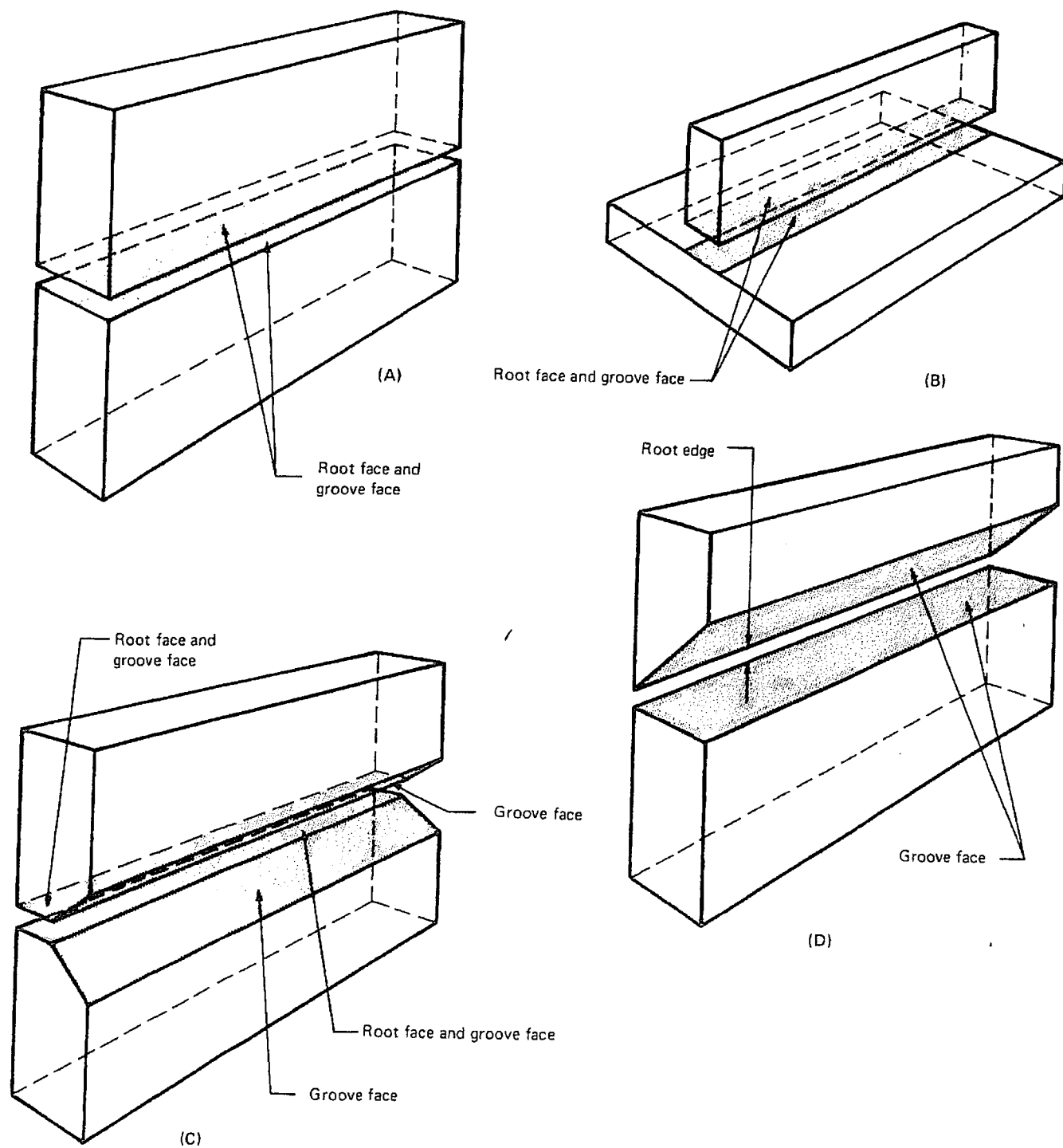
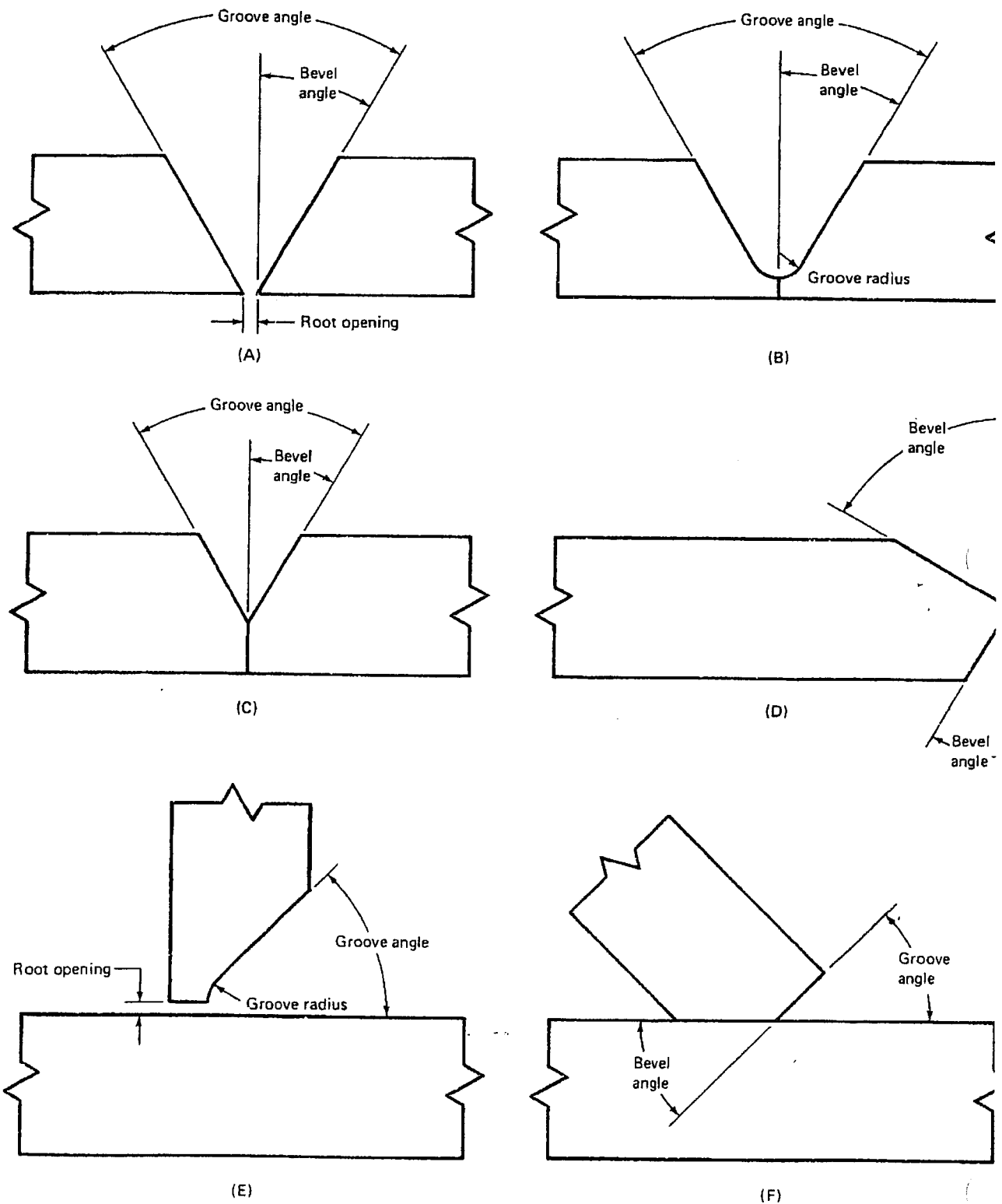
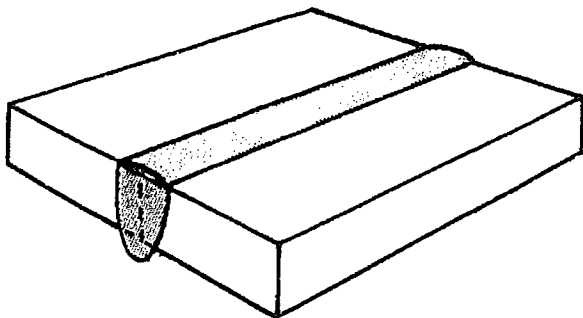


Fig. 3—Groove face, root edge, and root face

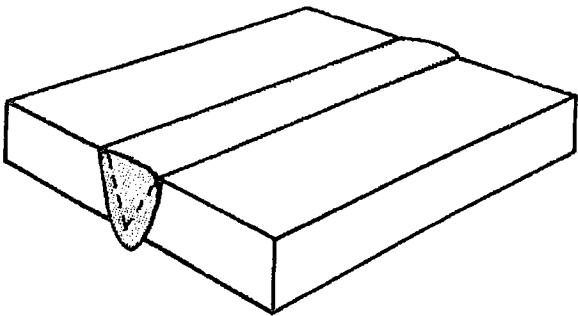


Additional terms that reference this figure bevel

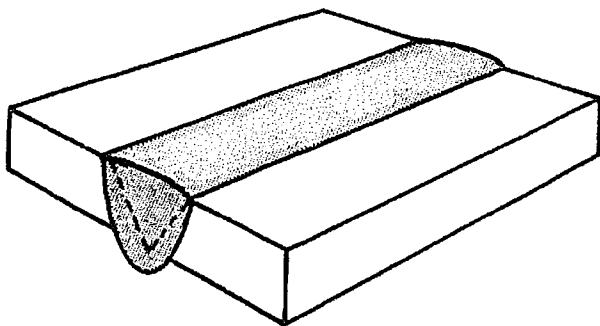
Fig. 4—Bevel angle, groove angle, groove radius, and root opening



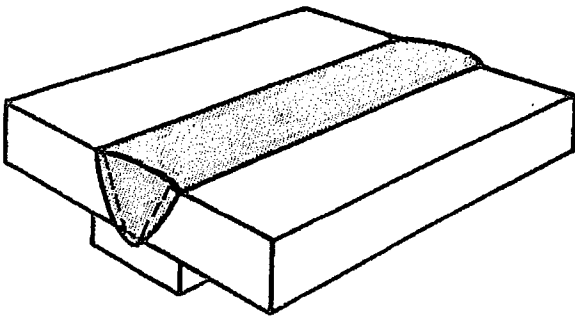
(A) Single-square-groove weld



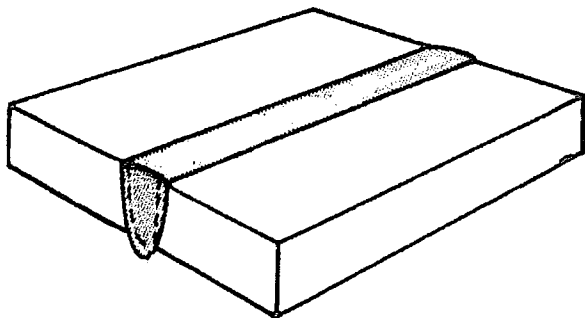
(B) Single-bevel-groove weld



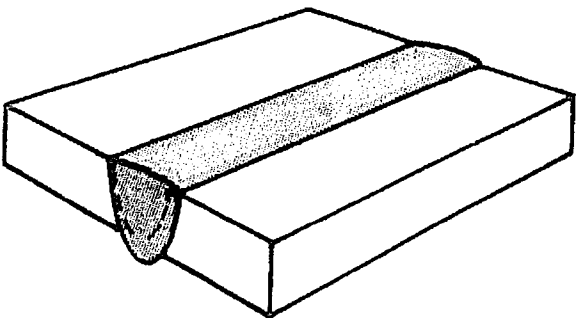
(C) Single-V-groove weld



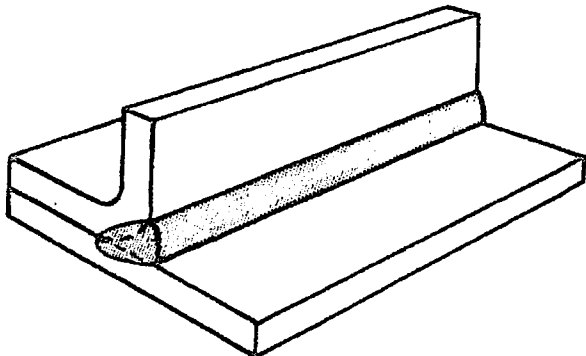
(D) Single-V-groove weld with backing



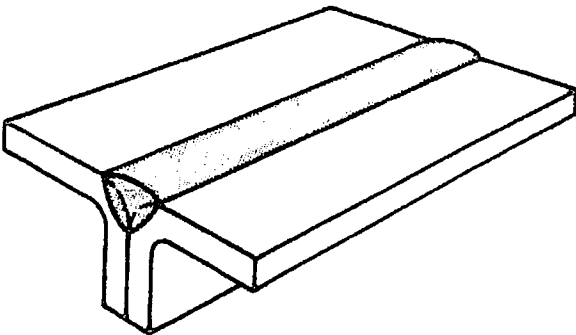
(E) Single-J-groove weld



(F) Single-U-groove weld



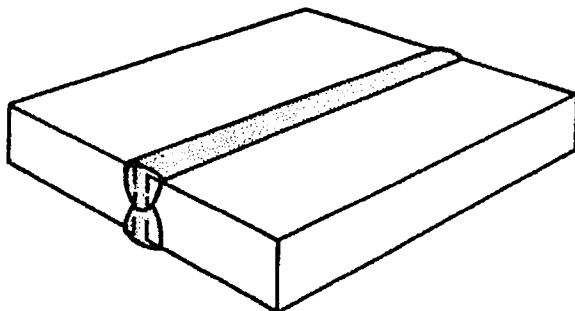
(G) Single-flare-bevel groove weld



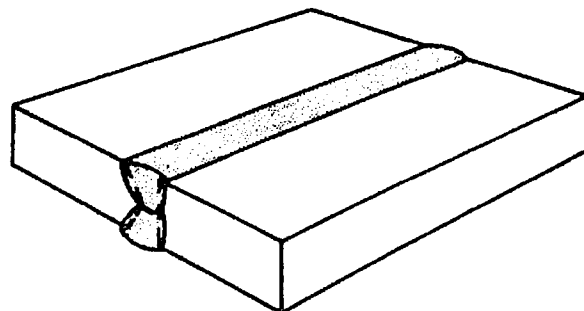
(H) Single-flare-V-groove weld

Additional terms that reference this figure
backing, groove weld, single-welded joint

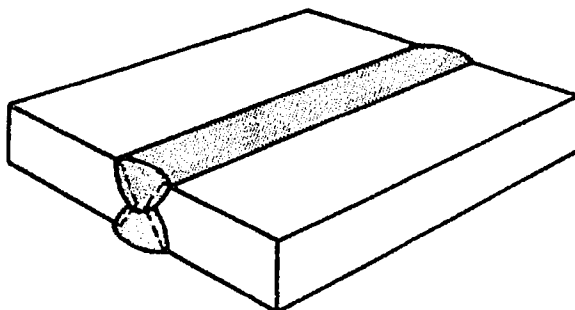
Fig. 5—Single-groove weld joints



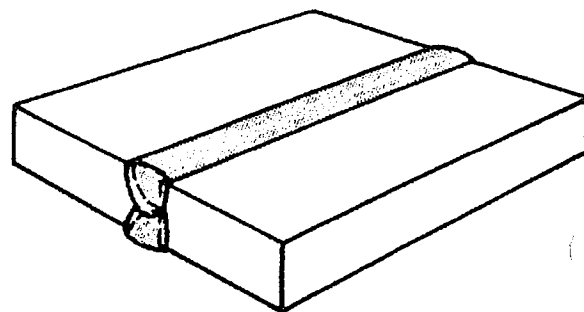
(A) Double-square-groove weld



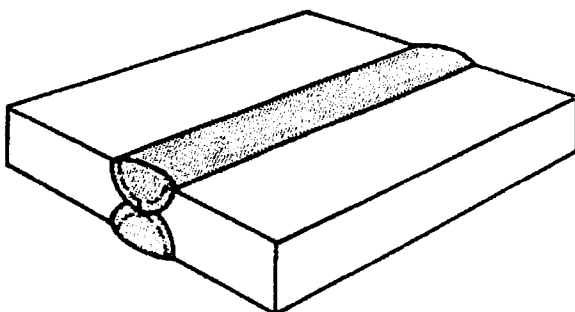
(B) Double-bevel-groove weld



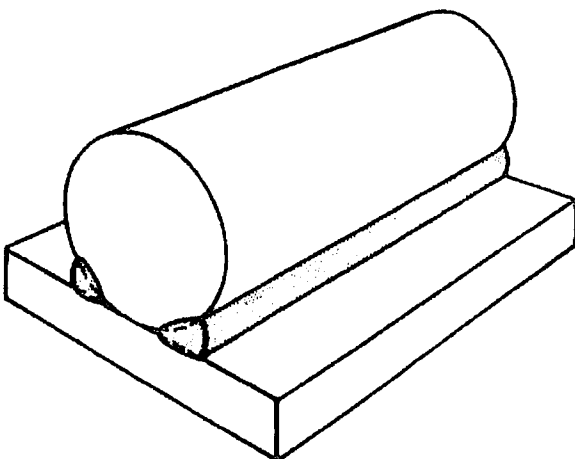
(C) Double-V-groove weld



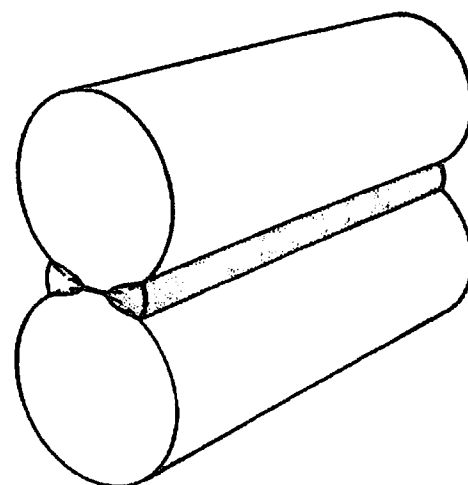
(D) Double-J-groove weld



(E) Double-U-groove weld



(F) Double-flare-bevel-groove weld

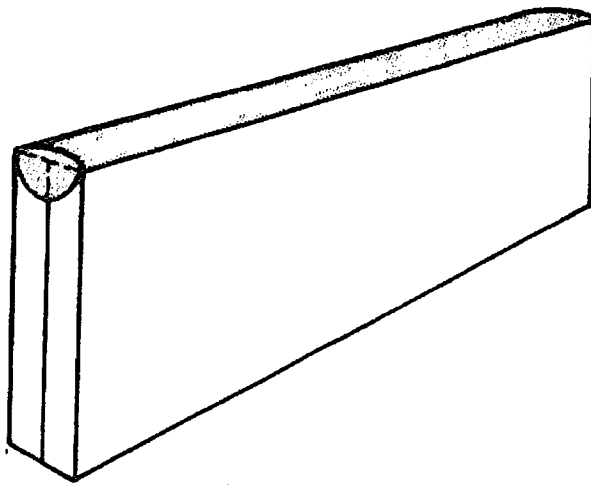


(G) Double-flare-V-groove weld

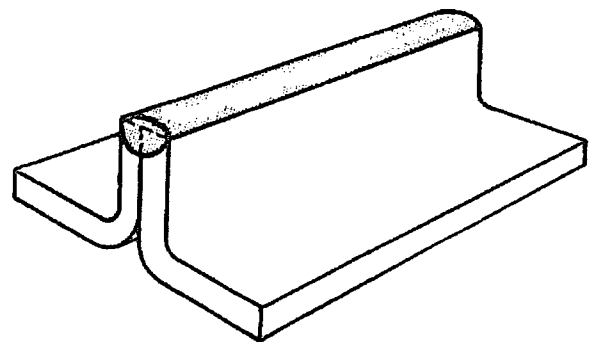
Additional terms that reference this figure
double-welded joint, groove weld

Fig. 6 — Double-groove weld joints

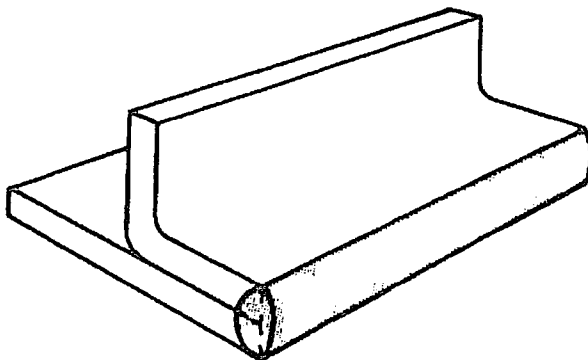
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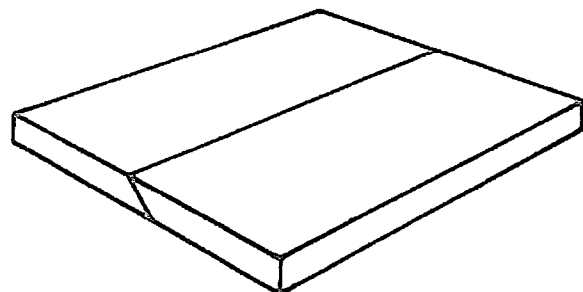
(A) Edge weld



(B) Edge-flange weld



(C) Corner-flange weld



(D) Scarf joint

Additional terms that reference this figure
flange weld

Fig. 7 — Edge weld, flange welds and scarf joint

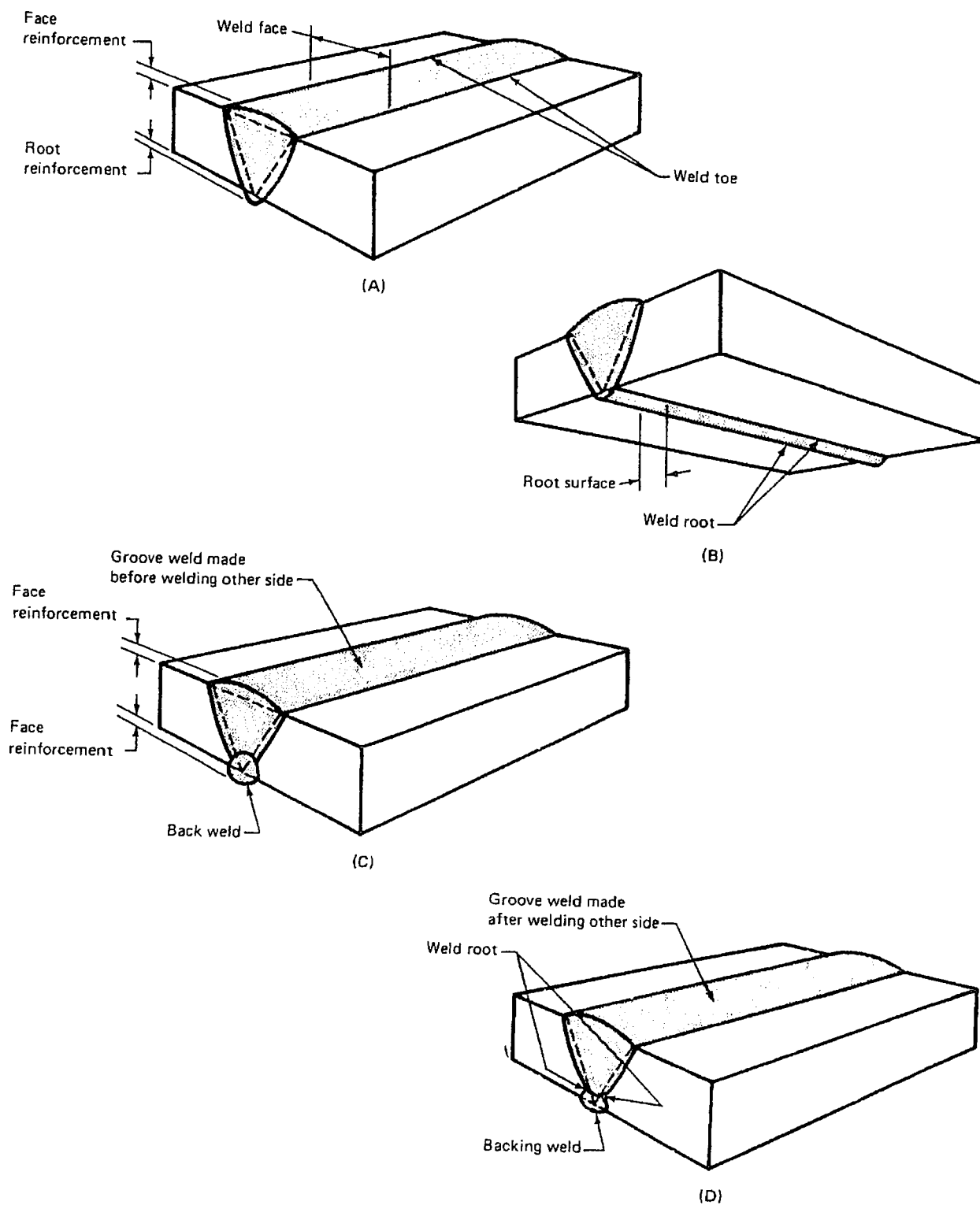
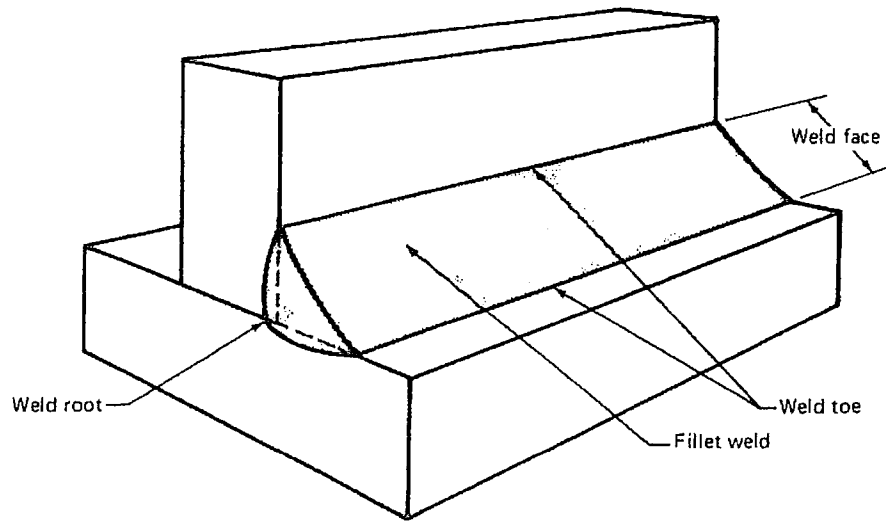
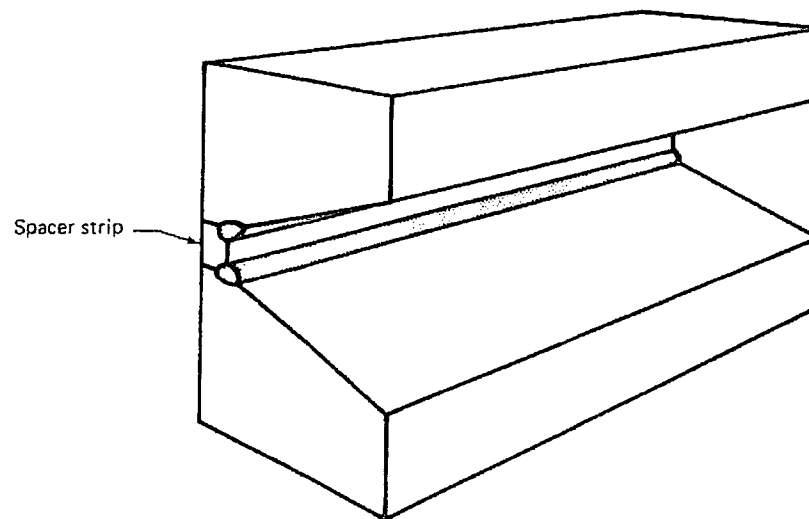


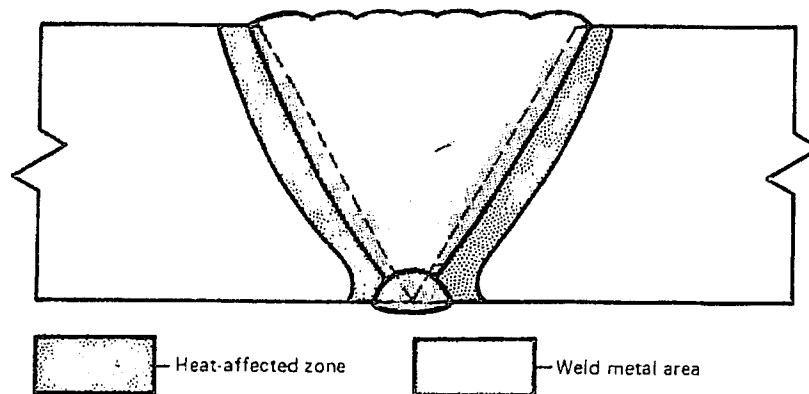
Fig. 17—Parts of a weld



(E)



(F)



(G)

Fig. 17 (cont.)—Parts of a weld

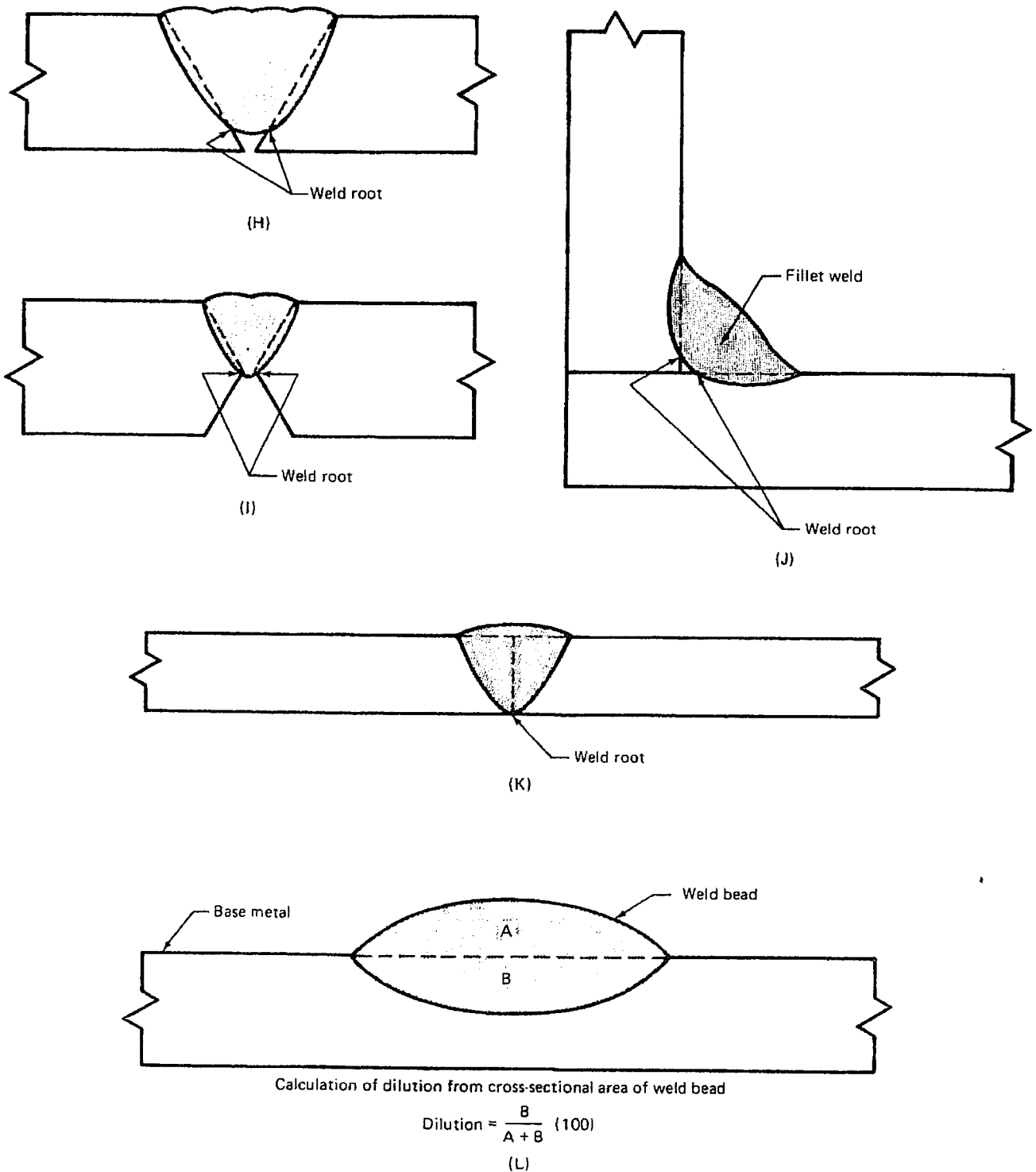
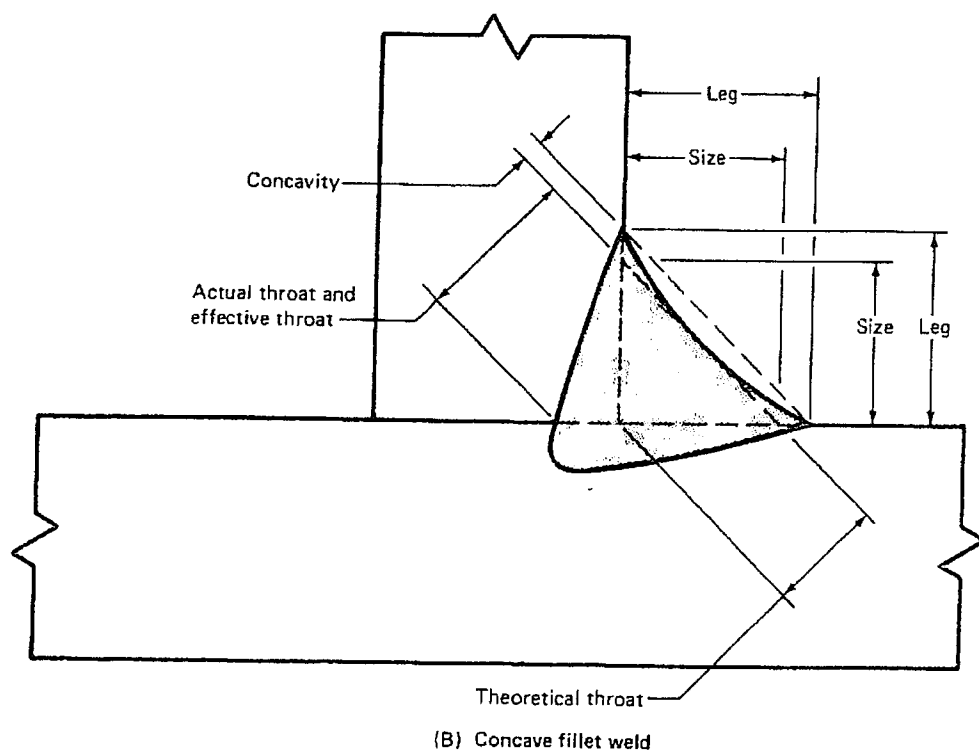
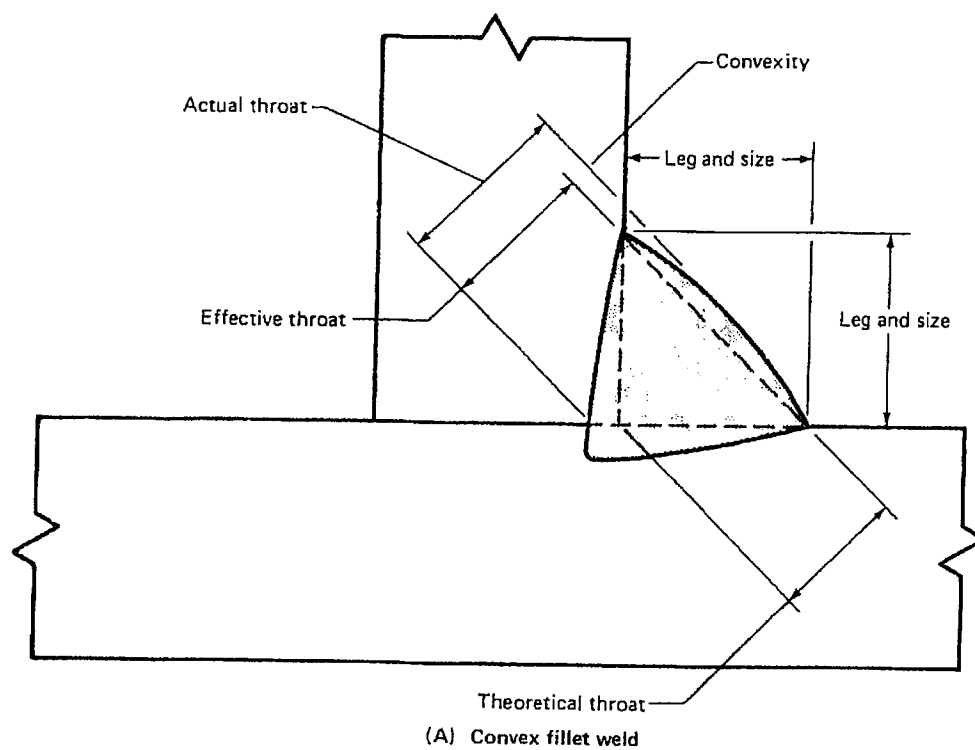


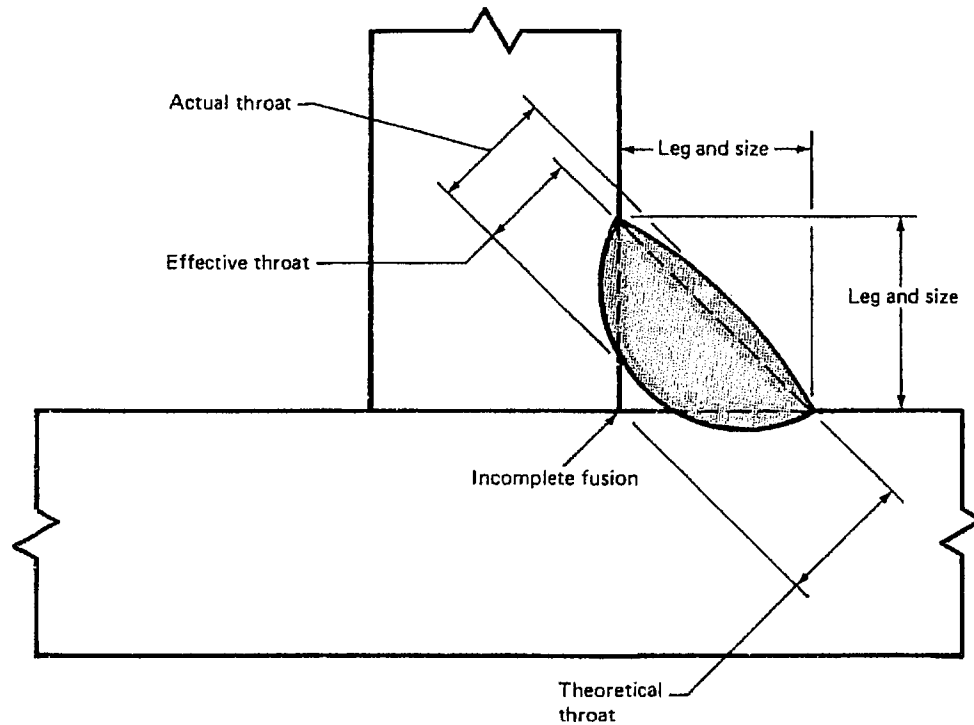
Fig. 17 (cont.)—Parts of a weld



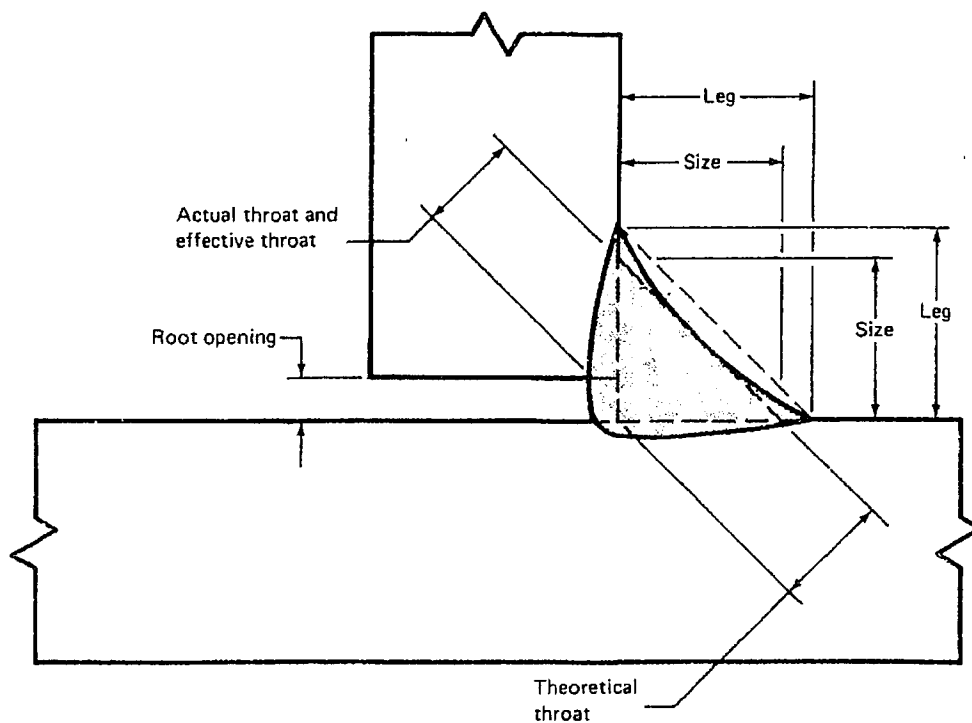
Additional terms that reference this figure
actual throat, effective throat, fillet weld leg,
fillet weld size, theoretical throat

Fig. 18—Weld sizes

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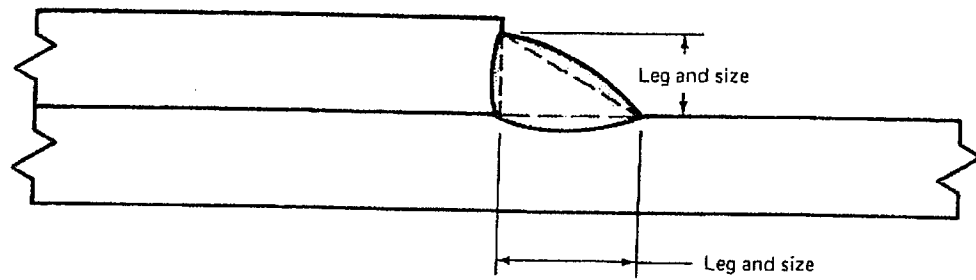


(C) Fillet weld with incomplete fusion

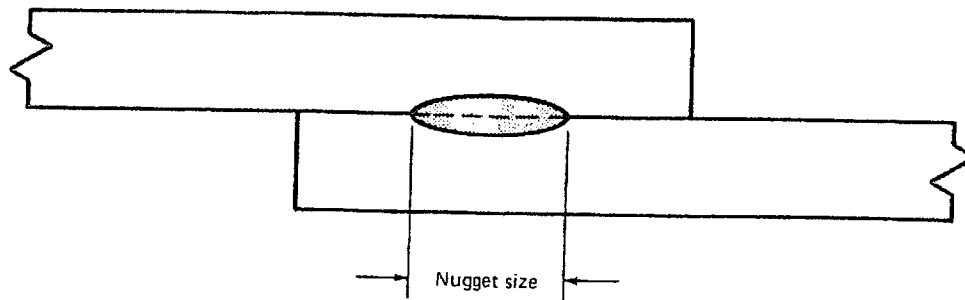


(D) T-joint with root opening

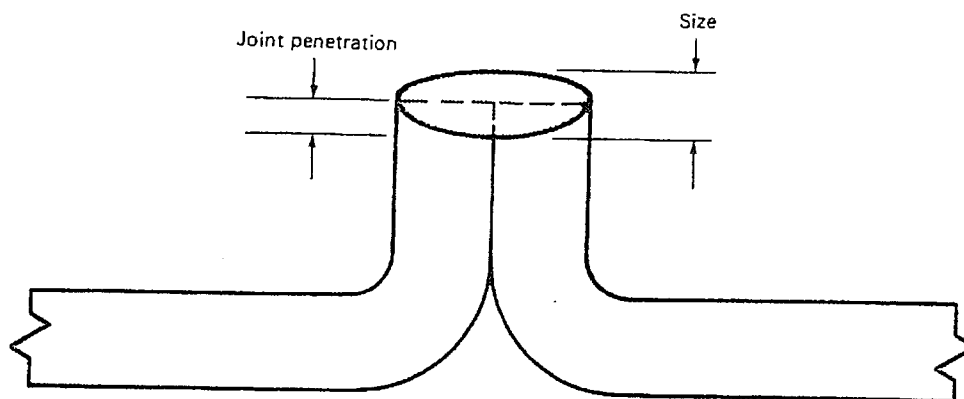
Fig. 18 (cont.)—Weld sizes



(E) Unequal leg fillet weld



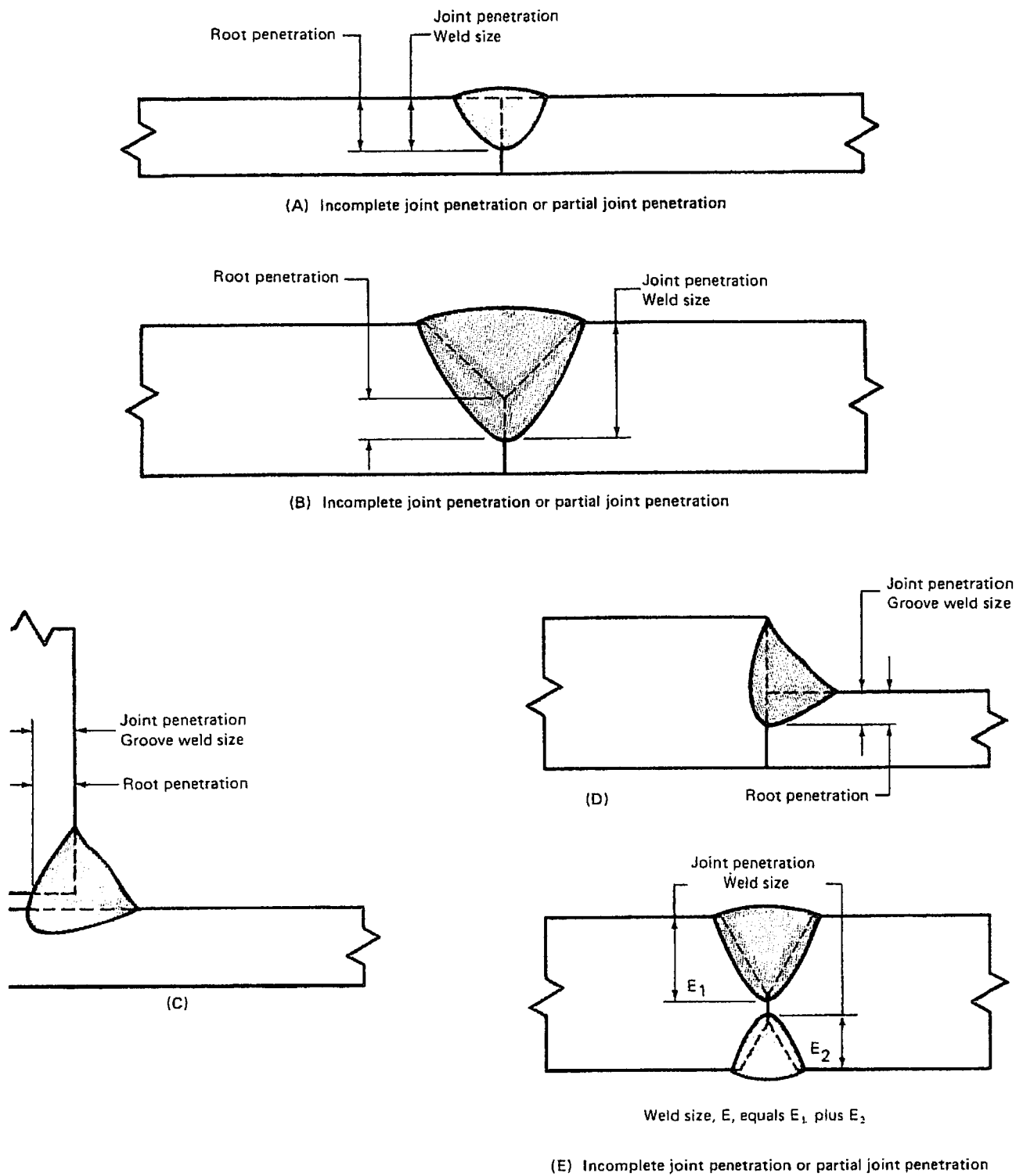
(F) Size of seam or spot weld



(G) Edge weld size or flange weld size

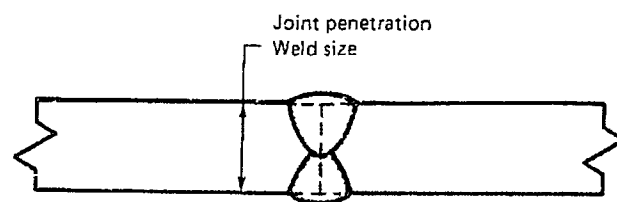
Fig. 18 (cont.)—Weld sizes

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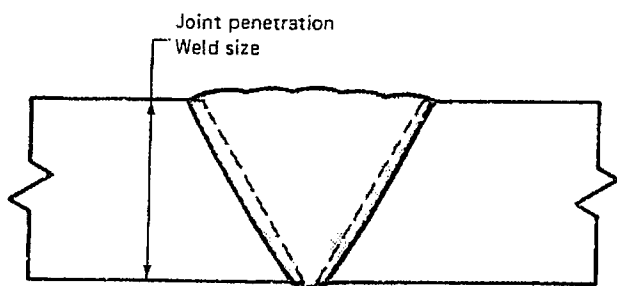


Additional terms that reference this figure
flange weld, root surface

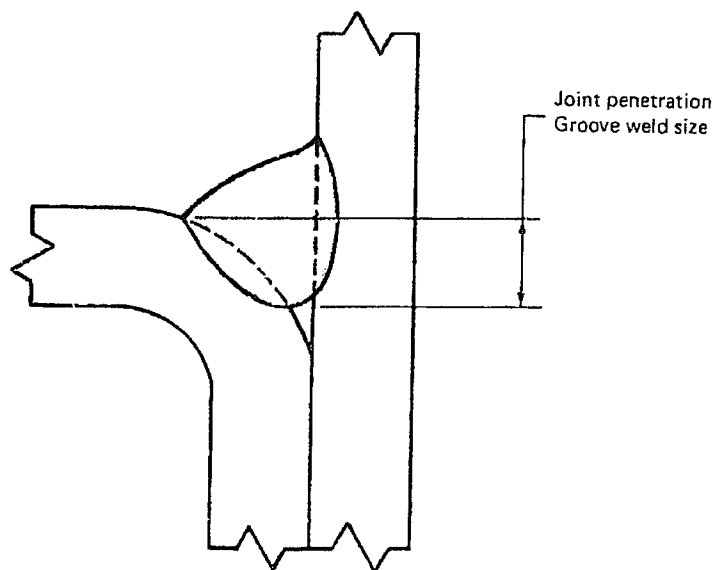
Fig. 19 — Joint penetration and root penetration



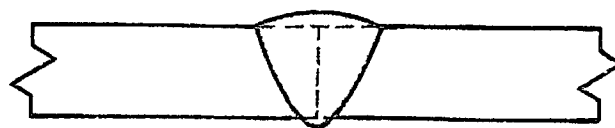
(F) Complete joint penetration



(G) Complete joint penetration



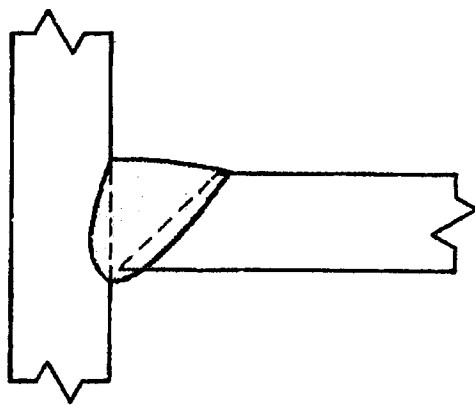
(H) Flare-bevel groove weld



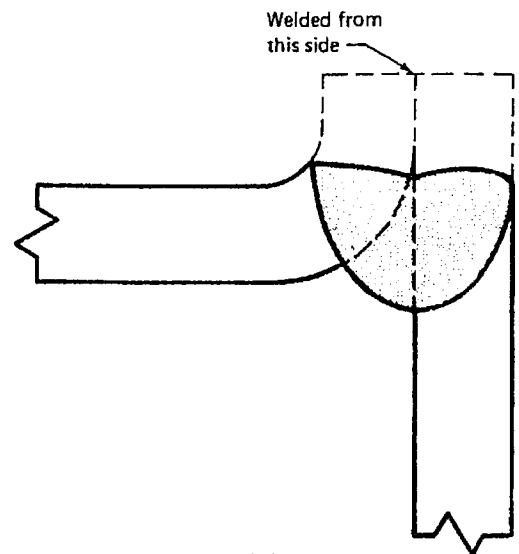
(I) Melt-through

Fig. 19 (cont.) — Joint penetration and root penetration

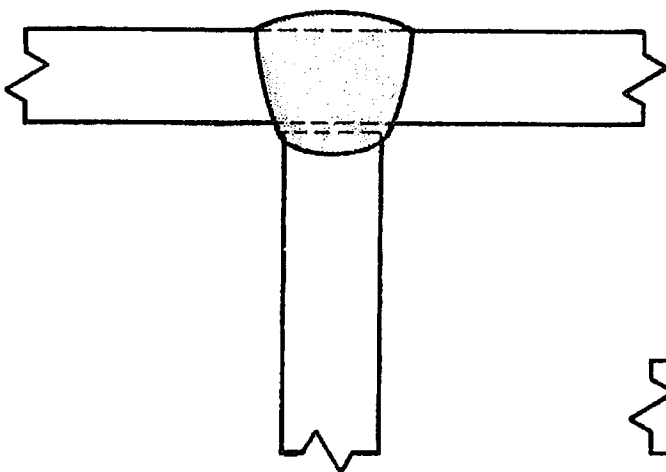
Figure 19/63



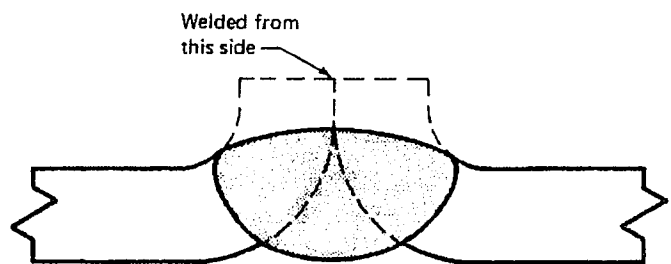
(J) Melt-through



(K) Melt-through*



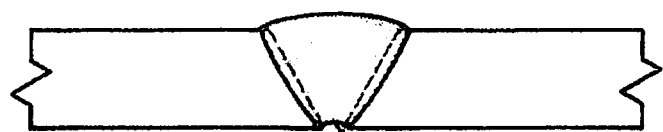
(L) Melt-through



(M) Melt-through*



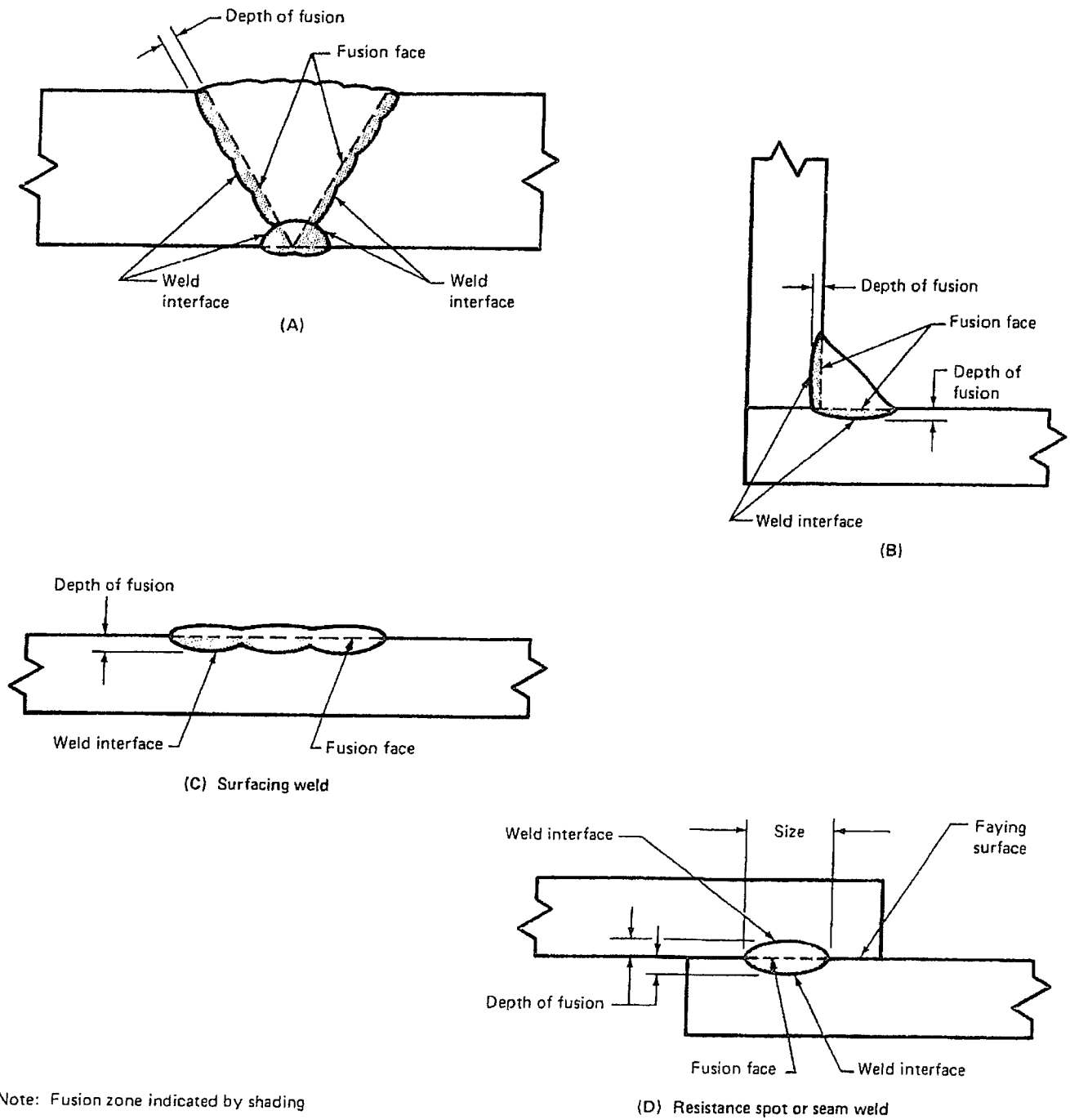
(N) Melt-through
Convex root surface



(P)
Concave root surface

*Special case of flange weld;
see also Figs. 7B and C.

Fig. 19 (cont.) — Joint penetration and root penetration



Additional terms that reference this figure
resistance seam welding, resistance spot welding

Fig. 22 — Fusion welds

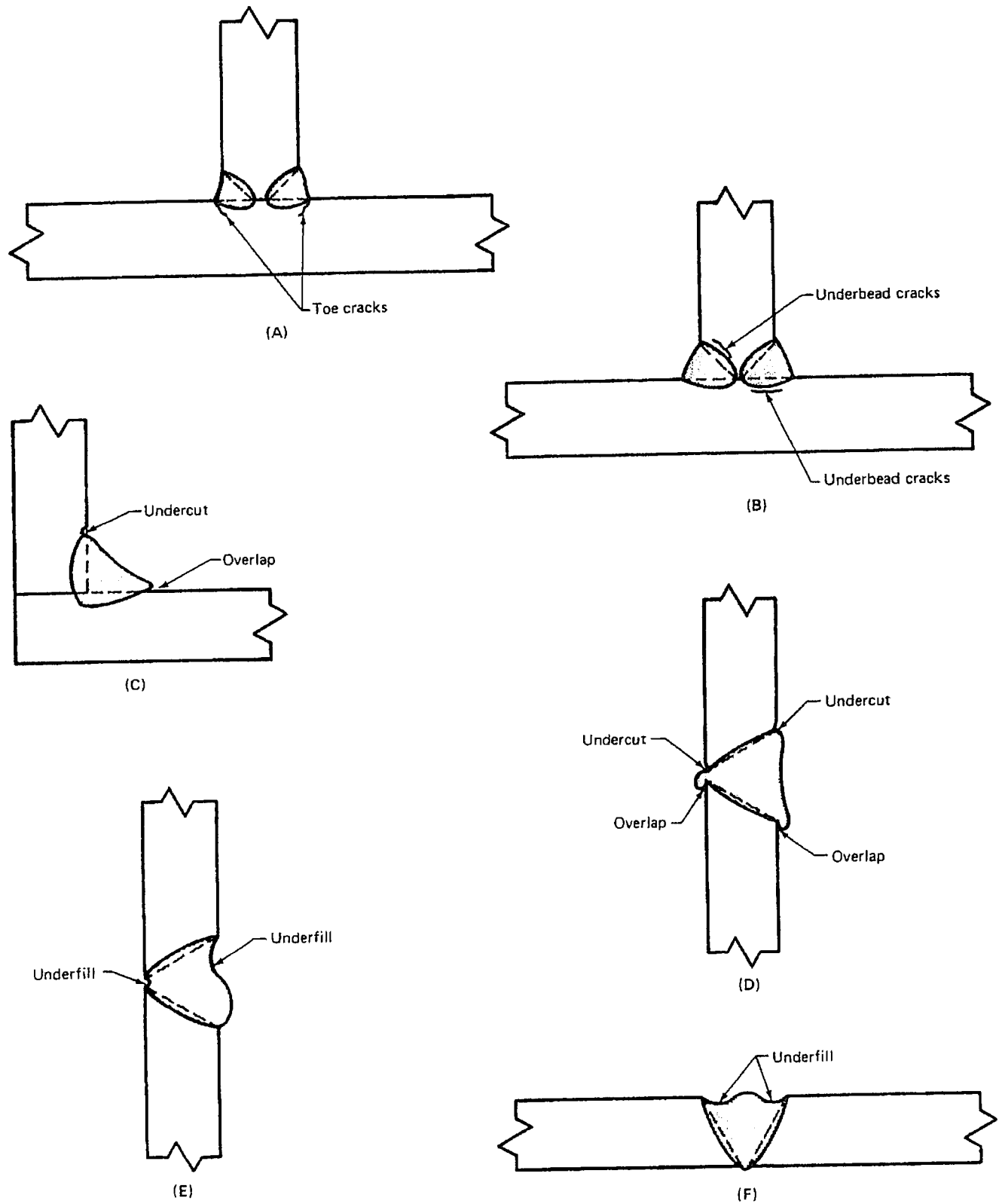
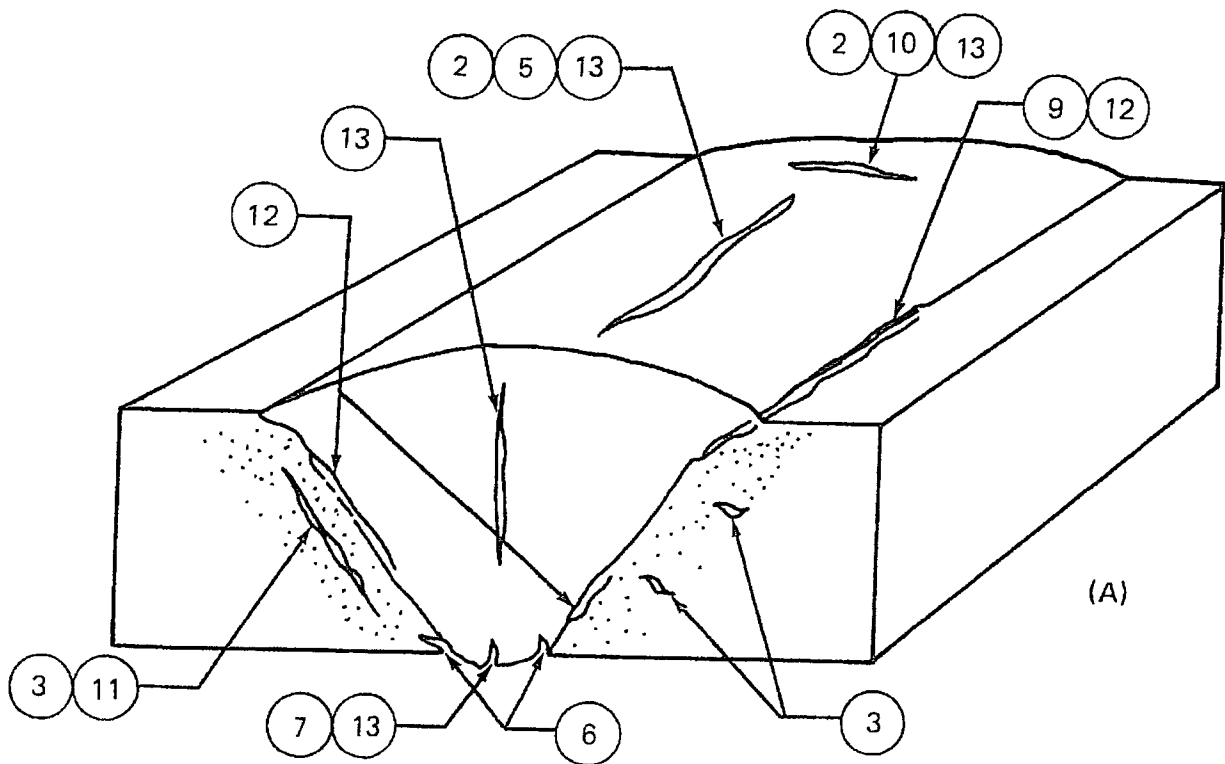
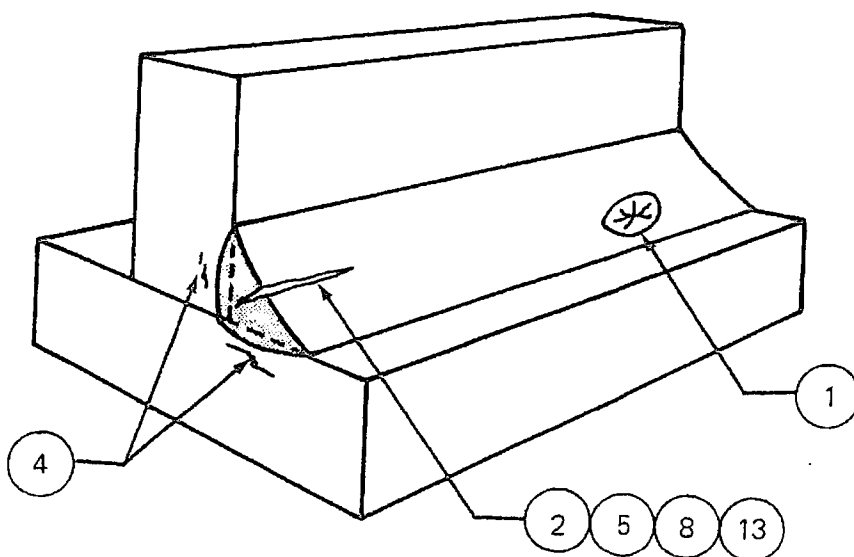


Fig. 24—Weld flaws



LEGEND

- 1 Crater crack
- 2 Face crack
- 3 Heat-affected zone crack
- 4 Lamellar tear
- 5 Longitudinal crack
- 6 Root crack
- 7 Root surface crack
- 8 Throat crack
- 9 Toe crack
- 10 Transverse crack
- 11 Underbead crack
- 12 Weld interface crack
- 13 Weld metal crack



(B)

Fig. 25—Crack types

